

Overview:

The current report consists of analyzing mortality rates using statistical models, such as the Gompertz model. The conclusions are drawn from the evaluation done in the R code.

1. Carrying out a Gompertz graduation of the death rates over the age range 25 to 80, using a model with binomial errors.

a) Plotting the observed death rates on the logistic scale against the ages x where $\hat{q}_x$ are the usual crude rate estimates at age x :

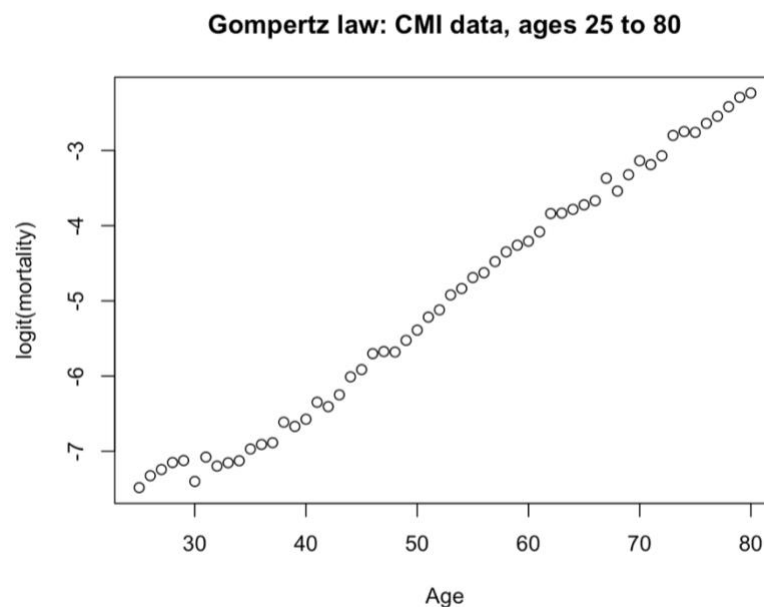


Figure 1: Observed Mortality Rates on Logit Scale by Age

Comment: The data appears to be linear, therefore following a Gompertz model's assumption, which states that the mortality rate increases in a geometric progression, hence, a straight line (Gompertz function) is obtained when death rates are plotted on a logarithmic scale. The plot appears to have some fluctuations in the younger ages as well as older ages, however the points appear well distributed without any significant deviations.

b) Fitting a Generalized Linear Model (GLM) with a linear predictor of the form $\text{logit}(q_x) = a_0 + a_1x$ to the crude rates $\hat{q}_x$, assuming a Gompertz law with binomial errors using the following line of code:

```
glm(Q.x ~ AGE, weights = E.init, family = binomial)
```

Obtaining the current output: Coefficients:

(Intercept)	AGE
-10.607	0.106

Degrees of Freedom: 55 Total (i.e. Null); 54 Residual

Null Deviance: 29610

Residual Deviance: 218.6 AIC: 634.5

c) From the results above, we can conclude that the expression for the fitted Gompertz line is:

$$\text{logit}(q_x) = -10.607 + 0.106 * \text{AGE}$$

where:

$a_0 = -10.607$ (intercept)

$a_1 = 0.106$ (slope)

d) Marking the fitted Gompertz line on the graph of Figure 1:

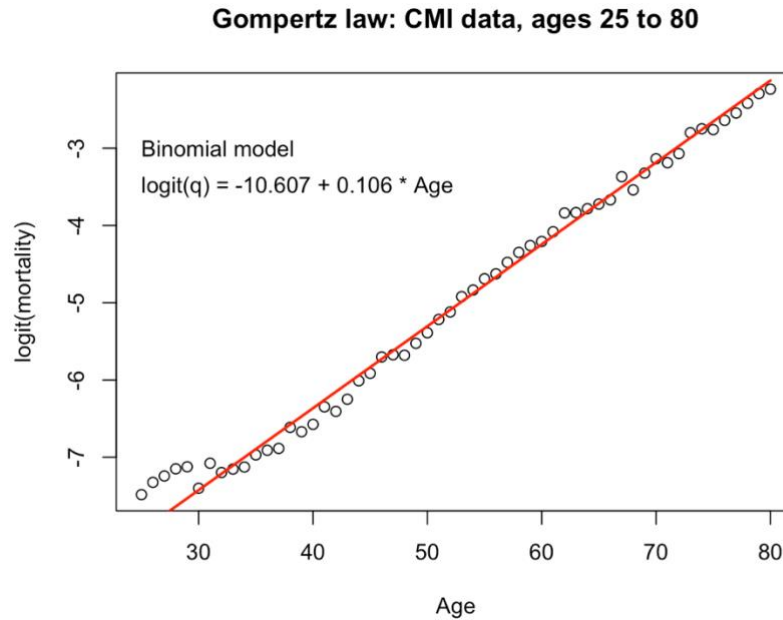


Figure 2: Observed Mortality Rates on Logit Scale by Age with a fitted Gompertz line

e) **Comment:** The fitted line (red) in Figure 2 indicates that the Gompertz model provides a proper approximation of the mortality, since it closely follows the data points that have been observed. As the plot in Figure 1, the current plot also contains some deviations at younger ages (25-40) which could be due to external factors that are unrelated to the age of an individual. It also contains some minor deviations at older ages (approx. 65-75). However, the model works better for the older ages, compared to younger.

f) Calculating and plotting the Pearson residuals of the fitted model:

The Pearson residuals can be calculated using the following formula:

$$z_i = \frac{d_i - \hat{d}_i}{\sqrt{\text{Var}(d_i)}} \approx \frac{d_i - \hat{d}_i}{\sqrt{\hat{d}_i}}$$

where:

$d_i \Rightarrow$ observed data

$\hat{d}_i \Rightarrow$ fitted data

Which could be evaluated through the following line of code:

```
resid(Gomp.bin, type="pear")
```



Figure 3: Pearson residuals in Binomial Gompertz model

g) **Comment:** The residuals appear to have a wavelike pattern strong in the residuals which indicates that the Gompertz model does not provide a reasonable fit for the mortality data. For the model to be a good fit, the residuals need to be randomly scattered. Some larger residuals are observed, particularly at younger and older ages, which could indicate some overdispersion, where the model underestimates the variability in mortality rates.

2. Fitting another Generalized Linear Model with binomial errors using a quadratic predictor having the following form:

$$b_0 + b_1(x - \bar{x}) + b_2(x - \bar{x})^2$$

a) First we use the following line of code to obtain the GLM:

```
glm(Q.x ~ I(AGE - mean(AGE)) + I((AGE - mean(AGE))^2), weights = E.init, family = binomial)
```

Therefore we obtain the outcome below:

```
Call: glm(formula = Q.x ~ I(AGE - mean(AGE)) + I((AGE - mean(AGE))^2),
family = binomial, weights = E.init)
```

Coefficients:

(Intercept)	I(AGE - mean(AGE))	I((AGE - mean(AGE))^2)
-5.0223535	0.1079472	-0.0001734

Degrees of Freedom: 55 Total (i.e. Null); 53 Residual

Null Deviance: 29610

Residual Deviance: 203.1 AIC: 621

The expression of the fitted quadratic equation of the model above is as follows:

$$-5.02235 + 0.1079(x - \bar{x}) - 0.00017(x - \bar{x})^2$$

where:

$$b_0 \Rightarrow -5.02235$$

$$b_1 \Rightarrow 0.1079$$

$$b_2 \Rightarrow -0.00017$$

- b) Using the plot() function in R, we obtain the same plot as is in Task 1, and another plot with an added Gompertz curve using:

```
lines(AGE, Gomp.quadratic$lin, lwd = 2, lty = 2, col = "red")
```

Obtaining the following plots:



Figure 4: Observed Mortality Rates on the LogitScale with a fitted Gompertz curve

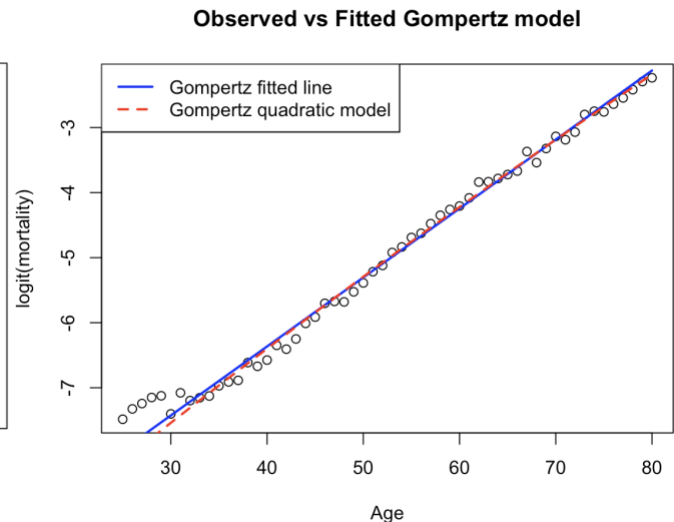


Figure 5: Fitted Gompertz line vs. Fitted Gompertz curve

- c) **Comment:** The quadratic Gompertz model in Figure 5, follows the linear model, with only slight deviations at younger ages, where it adjusts slightly for observed mortality trends. Both models fit the data for middle and older ages similarly, with some slight improvements in the younger ages through the quadratic model. Therefore, the quadratic model is a better fit for the younger population.

- d) Computing the value of the Akaike Information Criterion (AIC) of the fitted GLMs using the expression:

$$AIC = 2k - 2\ln(\hat{L})$$

where:

$\hat{L} \Rightarrow$ maximized value of the likelihood function for the model

$k \Rightarrow$ the number of estimated parameters in the model

Therefore, knowing the information above, we can create a function in R that uses the AIC formula And calculates the AIC value depending on the model:

```
AIC_calculation <- function(model) {
  k <- length(coef(model))
  log_Likelihood = as.numeric(logLik(model))

  AIC_output <- 2*k - 2*log_Likelihood #AIC = 2k - 2ln(MLE)

  return(AIC_output)
}
linear_model_AIC <- AIC_calculation(Gomp.bin) #2 parameters
quad_model_AIC <- AIC_calculation(Gomp.quadratic) #3 parameters
```

The AIC could also be calculated through a built in function in R, `AIC()`, which would result in the same output.

Hence, we obtain the values of the AIC for both models:

AIC of the linear model = 634.5361

AIC of the quadratic model = 620.9757

Comparing the AIC for the two models, it is evident that the quadratic model is a better fit, since the models with smaller AIC are usually preferred.

References

F79SU Survival Models: Course Notes, (2025), Heriot-Watt University